Sparse-Guided Partial Dense for Cross-Modal Remote Sensing Image–Text Retrieval

Zuopeng Zhao, Xiaoran Miao, Lei Liu, Xinzheng Xu, Ying Liu, Jianfeng Hu, Bingbing Min, Yumeng Gao, and Kanyaphakphachsorn Pharksuwan

***Abstract*— Cross-modal remote sensing image–text retrieval (CMRSITR) involves retrieving relevant samples in one modality based on a query from another modality. Previous dense retrieval methods utilizing multivector dense representations have significantly enhanced retrieval performance. Meanwhile, recent advances in sparse retrieval have demonstrated that sparse representations offer comparable performance with enhanced interpretability and faster retrieval speeds. However, effectively integrating the strengths of these two paradigms to enable efficient and accurate retrieval in large-scale remote sensing (RS) image–text datasets remains an open challenge. In this study, we propose sparse-guided partial dense (SGPD) cross-modal retrieval, a novel approach that efficiently transforms dense vectors from pretrained dense retrieval models into sparse representations and leverages the overlap between sparse retrieval results and dense vector clusters to achieve high-precision and fast retrieval. By probabilistically selecting a limited number of dense clusters containing top sparse results, SGPD ensures retrieval efficiency while minimizing both memory and time costs. Designed as a plug-and-play solution, SGPD can be seamlessly integrated into existing RS image–text retrieval (RSITR) models without requiring modifications to their architectures. Extensive experiments on RS image–text datasets of varying scales demonstrate that SGPD achieves retrieval accuracy comparable to dense retrieval methods while significantly reducing training time and memory consumption.**

***Index Terms*— Cross-modal remote sensing image–text retrieval (CMRSITR), dense retrieval, large-scale remote sensing (RS) image–text datasets, sparse retrieval.**

I. Introduction

IN RECENT years, the rapid advancement of remote sensing (RS) satellite technology has significantly enhanced both the volume and quality of RS data [1], [2]. However, efficiently retrieving this ever-growing repository of multimodal RS data has become a critical challenge in the field [3], [4]. Among the various retrieval techniques, image–text retrieval, a classical approach to cross-modal information retrieval, has gained increasing prominence in RS applications [5], [6]. The unique characteristics of RS data, including diverse modalities, complex scenarios, and varied tasks, present significant challenges for efficient and accurate data retrieval. As a result, developing an image–text retrieval model capable of effectively handling large-scale RS datasets, while addressing the specific complexities of RS scenarios, has become a pivotal research focus in the domain.

RS image–text retrieval (RSITR) datasets [7], [8] often contain thousands of RS images along with their corresponding captions. As these datasets continue to grow in size, ensuring fast response times and efficient retrieval while maintaining high accuracy has become a significant challenge. For example, the high-dimensional features of RS images and the diversity of land cover types make semantic alignment between images and texts inherently complex. Simultaneously, efficiently processing the vast amounts of data in these large datasets and reducing computational costs to prevent retrieval speed degradation due to the sheer volume of data remains a critical challenge in large-scale RSITR.

Current methods for RSITR primarily rely on two paradigms: dense retrieval [9], [10], [11] and sparse retrieval [12], [13], as illustrated in Fig. 1(a) and (b). Dense retrieval, which represents both images and texts as dense vectors, offers higher retrieval accuracy but comes with high computational costs. This is especially problematic for large-scale datasets, where the time and storage space required for dense retrieval can become prohibitive. In contrast, while sparse retrieval is more computationally efficient and can significantly speed up retrieval, it typically suffers from lower accuracy. On large-scale, complex RS datasets, sparse retrieval may fail to provide sufficient accuracy. Therefore, achieving an optimal balance between computational efficiency and accuracy in retrieval tasks for large-scale RS datasets remains a key challenge in current research.

In this study, we introduce a sparse-guided partial dense (SGPD) cross-modal retrieval paradigm for RSITR, as depicted in Fig. 1(c). Our approach begins by training a lightweight sparse projection (SP) head on dense vectors extracted from a pretrained model, leveraging image–text dense scores for supervision. By employing an efficient transformation strategy and an optimized indexing adjustment mechanism, this paradigm achieves significant improvements

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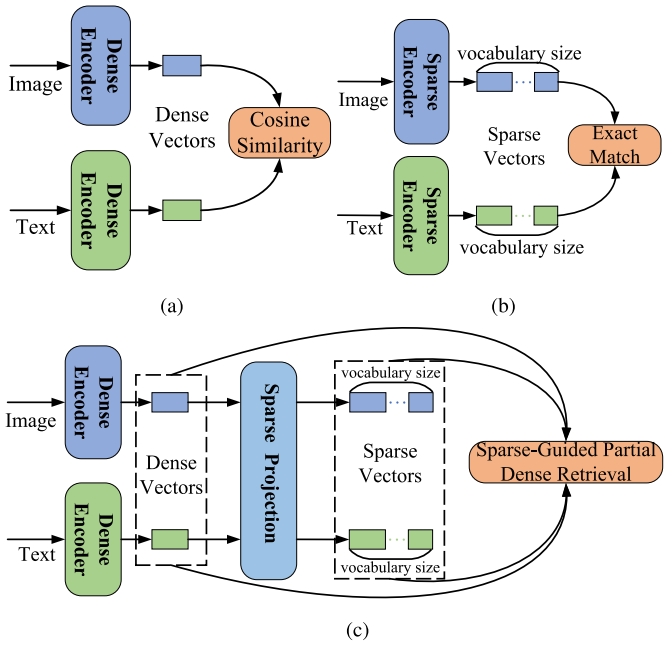


Fig. 1. Illustration of different image–text retrieval approaches. (a) Dense retrieval paradigm. (b) Sparse retrieval paradigm. (c) SGPD retrieval paradigm.

in both retrieval speed and accuracy. Additionally, we refine the fusion of sparse and dense retrieval results through clustering and probabilistic selection mechanisms. These techniques enable our method to deliver more relevant and efficient retrieval performance, even on large-scale RS image–text datasets.

Dense and sparse retrieval paradigms have been extensively studied and applied in single-modal text retrieval tasks [14], [15], [16]. However, in the context of multimodal RSITR, dense retrieval offers distinct advantages. These advantages arise primarily from the continuous nature of image data—composed of densely arranged pixel values—which poses significant challenges when mapping into a discrete vocabulary space due to inconsistencies in semantic segmentation and feature extraction. In contrast, dense retrieval utilizes vector representations to directly measure the similarity between multimodal features, effectively bridging the semantic gap between image and text modalities. Additionally, training a sparse retrieval model from scratch is a complex, multistep process involving the design of discretization features, vocabulary construction, and model optimization. Consequently, the training complexity and computational cost of sparse models are often comparable to, if not greater than, those of dense models.

To address the aforementioned challenges, we propose a novel approach that integrates the semantic richness of dense representations with the computational efficiency of sparse representations. Specifically, our method leverages a pretrained dense encoder model and introduces a lightweight projection layer that maps dense vectors to sparse vocabulary representations while preserving the semantic expressiveness of the original dense model. This design minimizes the additional training burden and retains the high-quality representation capabilities of dense encoders. To address issues such as high-dimensional co-activation and semantic bias [17] during the sparsification of image features, we propose a mechanism based on Bernoulli random variables and a parameterized mapper. By dynamically disabling redundant lexical extensions, this mechanism compels the projection layer to produce sparse vocabulary representations with clear semantics, thereby enhancing interpretability and semantic consistency. Furthermore, this approach eliminates the high dual-GPU resource requirements typically needed to train both dense and sparse models simultaneously, making it particularly well-suited for resource-constrained scenarios. In the sparsification process, while minimizing the ℓ_1 norm of representations is a common method to encourage sparsity, it does not ensure an even distribution of inverted index lists, which can negatively affect retrieval efficiency [18]. To resolve this issue, we introduce a FLOPs regularizer inspired by Zipf's law. This regularizer smooths and optimizes the computation of average floating-point operations required for scoring image or text queries, achieving a balanced trade-off between index sparsity and retrieval efficiency. By aligning with the fundamental objectives of sparse retrieval paradigms, this design significantly improves performance in real-world retrieval tasks.

Combining sparse and dense retrieval scores is a widely adopted approach in document retrieval [19], as these two paradigms capture complementary relevance signals, thereby enhancing retrieval effectiveness. However, independently executing these retrieval methods or relying on simple linear interpolation may fail to exploit their full potential. To address this limitation, we propose a method that begins by clustering dense vectors and then selecting the top-ranked results from sparse retrieval. Next, we calculate the overlap between these results and assign a priority score to each dense cluster. Using these scores, we probabilistically sample a limited number of clusters for further evaluation. As a result, instead of computing dense scores for all candidate samples, we focus on a constrained subset, significantly reducing retrieval time in large-scale datasets without compromising retrieval performance. Our codes are available at https://github.com/MXiaoran/SGPD.

The contributions of our work can be summarized as follows.

1) We propose an efficient sparse transformation strategy that converts dense vectors into sparse representations. This method significantly reduces floating-point computations while maintaining comparable accuracy, thereby saving storage space and improving computational efficiency for large-scale RSITR.
2) We introduce a novel SGPD cross-modal retrieval paradigm tailored for RSITR. This paradigm effectively addresses the challenges of large-scale datasets, enabling efficient and accurate retrieval.
3) We conducted extensive experiments on both small-scale and large-scale benchmark datasets. The experimental results demonstrate that our method balances recall and retrieval speed across datasets such as RSICD [7], RSITMD [8], UCM [20], and other large-scale datasets, effectively proving its robustness and scalability.

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The structure of this article is organized as follows. Section II provides a concise review of related work, focusing on cross-modal RSITR (CMRSITR), sparse retrieval, and its integration with dense retrieval. Section III details the proposed method and its underlying principles, offering a comprehensive explanation of our approach. Section IV presents extensive comparative and ablation experiments to validate the effectiveness of the proposed method. Finally, Section V concludes the article by summarizing our findings and contributions.

II. Related Work

A. Cross-Modal RSITR

The task of CMRSITR involves retrieving corresponding RS images based on semantic information extracted from descriptive caption. Early CMRSITR approaches can generally be categorized into two main types: caption-based methods [7], [20], [21], [22] and embedding-based methods [23], [24]. Caption-based retrieval methods typically rely on caption generators to create captions for each RS image in the database. Ni et al. [25] incorporated object counts into RS image captioning, as well as emerging areas such as underwater image captioning [26], [27]. During the retrieval phase, the similarity between the query text and the generated captions is computed, often using various evaluation metrics [28], [29], [30], [31]. Although two-stage caption-based CMRSITR methods have matured over time, they come with additional training costs and can amplify errors. Embedding-based, single-stage methods aim to directly measure similarity by aligning data from different modalities into a unified subspace [5], [32], [33]. While embedding-based methods have made progress in certain areas, they primarily focus on enhancing cross-modal feature alignment, often overlooking the unique distribution characteristics of RS data and the homogeneity of feature distributions in high-dimensional common spaces. Furthermore, these methods tend to neglect the complexity differences between different sample pairs.

Recent advancements in transformer-based multilingual models have significantly enhanced the performance of CMRSITR tasks, particularly through the use of pretrained CLIP models [34]. Vision-language pretraining (VLP) focuses on understanding the semantic relationships between visual and textual elements by pretraining on large-scale image–text pair datasets. Inspired by the success of transformer-based language models (e.g., BERT [35], XLNet [36], BART [37]) and Vision Transformers (ViT) [38], VLP has become a mainstream approach for learning multimodal representations. This approach has achieved notable success in downstream tasks such as image captioning [39], image–text retrieval [40], and visual question answering (VQA) [41]. Current VLP research can be categorized into two main types: single-stream and dual-stream models, based on their structural design. Single-stream models [42], [43] combine text and visual features before processing them through a shared transformer encoder. This configuration is more parameter-efficient, as it uses a single set of shared parameters to handle both modalities. However, since similarity scores must be computed for all possible image–text combinations, these models often experience slower retrieval speeds during inference. In contrast, dual-stream models [34], [44], [45] use separate encoders to extract text and visual features independently without sharing parameters. While these models perform well in image–text retrieval tasks, they are less effective at capturing complex interactions between vision and language in other multimodal understanding tasks.

The application of pretrained models in RS image research has seen significant advancements [46]. Zhang et al. [47] introduced a transfer learning approach from natural scenes to the RS domain, achieving impressive results in tasks such as scene classification, object detection, and land cover classification. Yuan et al. [10] proposed a parameter-efficient transfer learning (PETL) method based on the pretrained CLIP model, focusing on effectively and efficiently transferring vision-language (VL) knowledge from the natural domain to the CMRSITR task. Zhao et al. [9] extended the CLIP model by incorporating masking techniques for both images and texts to capture more fine-grained features. Additionally, they introduced and maintained two dynamically updated negative sample queues, which significantly improved retrieval performance. Liu et al. [11] introduced the first foundational VL model for the RS domain, designed to learn robust visual features with rich semantics and aligned text embeddings. This model not only excels in image–text retrieval tasks but also facilitates seamless integration with downstream applications, such as zero-shot image classification, linear probing, and k-NN classification.

B. Learned Sparse Retrieval

Sparse retrieval originated in document retrieval [48], [49], where queries and documents are encoded as sparse lexical vectors that can be efficiently indexed using an inverted index. Recently, several studies [12] have attempted to extend sparse retrieval from unimodal text retrieval to multimodal image–text retrieval. Lexlip [13] introduced a lexicon-bottlenecked language image pretraining, constructing a continuous bag-of-words bottleneck and learning the distribution of lexical importance using a lexical bottleneck module situated between a dual-stream encoder and a simplified text decoder. Lsr [17] employs Bernoulli random variables to control query expansion, addressing issues of high-dimensional co-activation and semantic bias in the sparse retrieval process. Sparse semantic features, such as those in the bag-of-words model, are more interpretable and require less retrieval time, but they are generally considered less accurate compared to dense representations.

Integrating the advantages of sparse and dense retrieval to achieve more efficient image–text retrieval is a topic of significant interest. In the field of unimodal text retrieval, various attempts [14], [50] have successfully balanced retrieval accuracy and speed. However, similar research is lacking in the domain of multimodal image–text retrieval. To address this gap, we propose a novel method that aims to effectively integrate sparse and dense representations, leveraging the efficiency of sparse representations while capitalizing on the feature-capturing strengths of dense representations.

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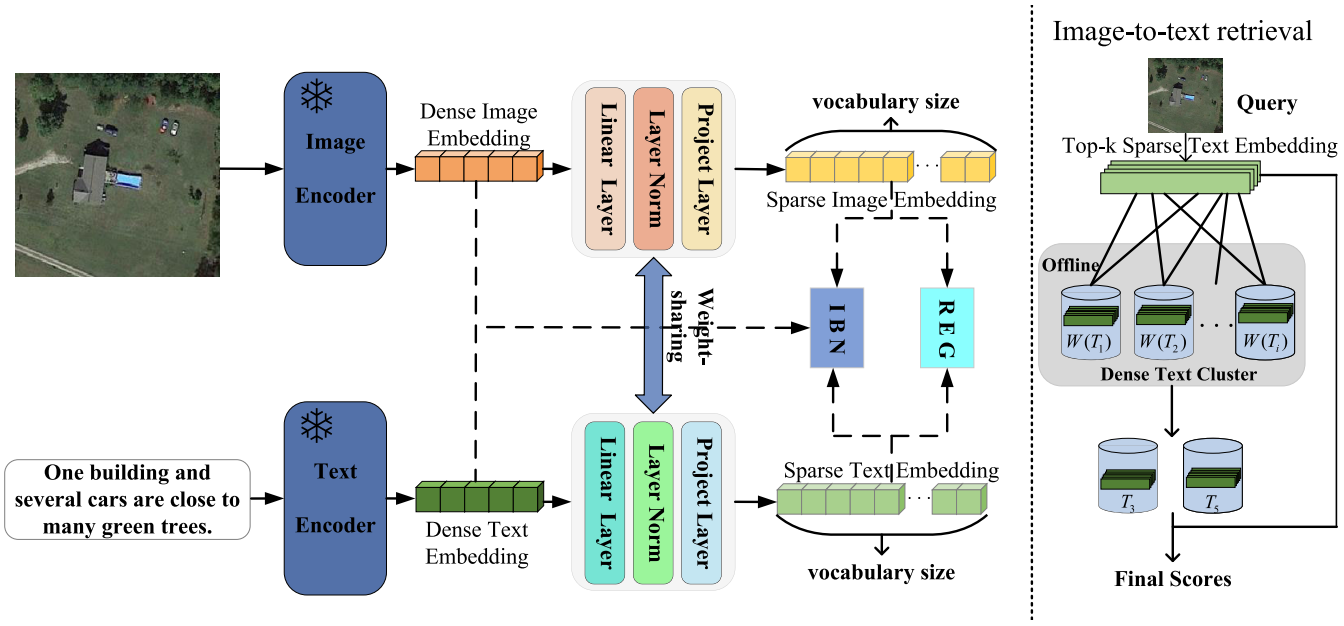


Fig. 2. Overview of the SGPD cross-modal retrieval. (Left) Dashed line illustrates the sparse retrieval process, where frozen dense vectors are converted into sparse lexical representations via a SP layer. (Right) Image–text retrieval task as an example, the process of performing dense retrieval based on the scores of a limited number of dense clusters is illustrated. Features from the sparse representations are matched within the dense clusters, ultimately resulting in the computation of the retrieval score.

By comprehensively utilizing both types of representations, our goal is to maintain retrieval accuracy while further enhancing retrieval speed.

III. Method

This section will introduce the SGPD framework we proposed. Fig. 2 presents an overview of SGPD, with specific details to be discussed in Sections III-A–III-D.

A. Preliminaries

Given an image–text pair dataset $\{(I_i, T_i)\}_{i=1}^{N}$, where (I_n, T_n) denotes the correspondence between image I_n and text T_n. The dense image encoder maps each image I_n to a latent space denoted as $I_n^d \in \mathbb{R}^d$. Similarly, the dense text encoder maps T_n to a latent space denoted $T_n^d \in \mathbb{R}^d$. Dense image and text encoders are utilized in cross-modal image–text retrieval, with notable examples including ALBEF [51] and BLIP [52]. Each dense encoder remains frozen during the retrieval process.

Taking the text as an example, the dense text vector of d-dimensional T_n^d is projected onto a ω-dimensional vector via $T_n^{d_1} = W_1 T_n^d$, where $W_1 \in \mathbb{R}^{w \times d}$ and $T_n^{d_1} \in \mathbb{R}^w$. Subsequently, layer normalization is performed

$$T_n^{d_2} = \gamma \times \frac{T_n^{d_1} - E\left(T_n^{d_1}\right)}{\sqrt{\mathrm{Var}\left(T_n^{d_1}\right) + \varepsilon}} + \beta \tag{1}$$

where $E(T_n^{d_1})$ and $\mathrm{Var}(T_n^{d_1})$ denote the expectation and variance of $T_n^{d_1}$, respectively. γ and β are the learnable affine transformation parameters, resulting in $T_n^{d_2} \in \mathbb{R}^w$. $T_n^{d_2}$ is then projected onto the word embedding space $C = \mathbb{R}^{|V|}$

$$T_n^c = \text{LM-Head}\left(T_n^{d_2}\right) \tag{2}$$

where $|V|$ represents the size of the vocabulary, and LM-Head is initialized similar to the word embedding of the transformer language model. Finally, we follow the previous approach to remove negative weights and reapply the normalization transform, resulting in T_n in the high-dimensional vocabulary space

$$T_n^p = \log\left(1 + \max\left(0, T_n^c\right)\right) \tag{3}$$

where $\max(\cdot, 0)$ ensures that all values are nonnegative to meet the sparsity requirement. The resulting $T^p \in \mathbb{R}^{|V|}$ represents the weights of the corresponding words in each dimension. Similarly, the same operation is applied to the images, resulting in $I^p \in \mathbb{R}^{|V|}$, representing the sparse representation of the words weighted images. Note that the parameters of the sparsification process are shared between text and images.

Additionally, we follow the D2S [17] probabilistic expansion control scheme to suppress high-dimensional co-activation and semantic deviation that may occur during the sparsification process. We use a dual-level probabilistic control framework comprising caption-level expansion and word-level expansion. At the caption level, a Bernoulli random variable $E_c \sim \mathrm{Ber}(p_c)$ is introduced to determine whether to expand the entire text caption, where $E_c = 1$ enables expansion and $E_c = 0$ prohibits it. For word-level control, each vocabulary word v_i is assigned a Bernoulli variable $E_i^v \sim \mathrm{Ber}(p_i^v)$ to regulate its expansion probability. During the initial training phase, the caption-level expansion probability p_c is set to zero, ensuring that the model exclusively learns mappings between images and their original textual captions. This strategy mitigates noise interference and reinforces precise cross-modal alignment. Word-level expansion probabilities p_i^v are initialized based on document frequency (df_i^v) of each word, formulated as $1 - df_i^v$. This design prioritizes expansion for low-frequency words (carrying rare semantic

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information) while suppressing expansion for high-frequency functional words (e.g., articles or prepositions) that exhibit generic semantics.

A progressive relaxation strategy is employed to dynamically adjust expansion probabilities during training. Specifically, a linear scheduler incrementally increases both caption-level and word-level probabilities across epochs. The caption-level probability follows:

$$p_c(t) = \frac{t}{T} \tag{4}$$

where t denotes the current epoch and T is the total number of epochs. For word-level control, the probability updates adhere to

$$p_i^v(t) = \min\left(p_i^v(t-1) + \frac{1 - df_i^v}{T}, 1\right). \tag{5}$$

Ensuring monotonic growth until saturation. This mechanism balances semantic diversity and stability throughout the training process.

B. Learning Sparse Representations

Traditional sparse representations for the first-stage retrieval are achieved through regularization ℓ_1. However, [18] points out that minimizing the ℓ_1 norm of the representation does not produce the most efficient index. Due to the Zipfian nature of the word frequency distribution, [18] introduces the flops regularizer to achieve a well-balanced index, which is a smooth relaxation of the average floating-point operations. This is essential to reduce the computation time required for querying and retrieving the corresponding images or texts. Specifically, considering image-to-text retrieval as an example

$$\ell_{\text{Flops}} = \sum_{j \in |V|} \left(\frac{1}{N} \sum_{n=1}^{N} T_{n,j}^p\right)^2 \tag{6}$$

where $T^p \in \mathbb{R}^{|V|}$, $T_{n,j}$ denotes the weight of token j in the nth text, and N is the batch size. The regularization loss for image–text can be defined as

$$\mathcal{L}_{\text{reg}}^{I2T} = \lambda_q \mathcal{L}_{\text{reg}}^{I} + \lambda_d \mathcal{L}_{\text{reg}}^{T} \tag{7}$$

where $\mathcal{L}_{\text{reg}}$ is the sparse FLOPs regularization (FR) in (4), and λ_q and λ_d are the regularization weights for the query and corresponding modals, respectively. Similarly, the regularization loss for text-to-image is and the total regularization loss

$$\mathcal{L}_{\text{reg}}^{T2I} = \lambda_q \mathcal{L}_{\text{reg}}^{T} + \lambda_d \mathcal{L}_{\text{reg}}^{I} \tag{8}$$

$$\mathcal{L}_{\text{reg}} = \left(\mathcal{L}_{\text{reg}}^{I2T} + \mathcal{L}_{\text{reg}}^{T2I}\right)/2. \tag{9}$$

C. Training Objective

Let $S(I_n^p, T_n^p)$ and $S(T_n^p, I_n^p)$ represent the ranking scores obtained by the sparse representations of image-to-text and text-to-image through (3). Taking image-to-text retrieval as an example, given a query image I_i^p, a positive text T_i^{p+}, a hard negative text T_i^{p-} (from frozen dense retrieval model), and a set of negative samples T_j^{p-} in the batch, we adopt the in-batch negatives sampling strategy to maximize the probability that the positive text T_i^{p+} is retrieved within the batch based on its relevance to the image

$$\mathcal{L}_{\text{IBN}}^{I2T} = -\log \frac{e^{S(I_i^p, T_i^{p+})}}{e^{S(I_i^p, T_i^{p+})} + e^{S(I_i^p, T_i^{p-})} + \sum_j e^{S\left(I_i^p, T_{i,j}^{p-}\right)}} \tag{10}$$

$$\mathcal{L}_{\text{IBN}}^{T2I} = -\log \frac{e^{S(T_i^p, I_i^{p+})}}{e^{S(T_i^p, I_i^{p+})} + e^{S(T_i^p, I_i^{p-})} + \sum_j e^{S\left(T_i^p, I_{i,j}^{p-}\right)}}. \tag{11}$$

Therefore, the total IBN loss is

$$\mathcal{L}_{\text{IBN}} = \left(\mathcal{L}_{\text{IBN}}^{I2T} + \mathcal{L}_{\text{IBN}}^{T2I}\right)/2. \tag{12}$$

By simultaneously optimizing the IBN loss and the regularization loss in (7), combining the advantages of both, the total loss can be defined as

$$L = \alpha L_{\text{IBN}} + (1 - \alpha)\lambda L_{\text{reg}} \tag{13}$$

where $\alpha = [0, 1]$ is a scalar weight and $\lambda = [0, 1]$ is a sparsity regularization parameter.

D. SGPD Retrieval

For the input image-text pair (I_n, T_n), we obtain dense vector representations (I_n^d, T_n^d) and sparse vectors (I_n^p, T_n^p) through the aforementioned model. For the sparse ranking scores, we re-rank and combine the results from both dense and sparse vectors. It is a natural approach to use the top-ranked results to guide the retrieval of certain dense vectors. However, current research indicates that simple re-ranking does not improve recall. Therefore, we employ the CDFS [14] method to expand and select the top sparse results and examine the distribution of the top-ranked results within the clusters. This allows us to include multiple top cluster results and adjust the evaluation weights to improve recall.

Specifically, for vector collections with a large number of high-dimensional embeddings that cannot be fully loaded into memory, we use a variant of binary k-means [53], which recursively divides large clusters into smaller ones. During the retrieval process, we first perform an initial sparse retrieval to obtain the top-k results. As shown in Algorithm 1, taking image-to-text retrieval as an example, with the query image I_i, we perform weighted overlapping on the sparse texts retrieved in the top-k and assign a priority score $W(T_c)$ to each text cluster T_c

$$W(T_c) = \sum_{T_{i,j} \in \mathcal{S}_i} \frac{S\left(I_i^p, T_{i,j}^p\right)}{\log\left(\text{Rank}S(T_{i,j}) + 1\right)} \tag{14}$$

where $\mathcal{S}_i = T_c \cap \text{TOPS}(k)$, $\text{TOPS}(k)$ is the set of top-k sparse retrieval results, $S(I_i^p, T_{i,j}^p)$ is the sparse ranking score from image I_i to text $T_{i,j}$, and $\text{Rank}S(T_{i,j})$ is the ranking position of text $T_{i,j}$ in $\text{TOPS}(k)$.

Specifically, for each image query I_i, we have a set of candidate texts $T_{i,j}$, with their matching scores $S(I_i, T_{i,j})$ obtained through sparse retrieval. To determine the threshold φ for each position K, we collect all Kth matching scores across multiple sample queries I_q

$$S(k) = \left\{S(I_i, T_{i,j(k)}) | i = 1, 2, \ldots, m\right\}. \tag{15}$$

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Algorithm 1 SGPD for Image-to-Text Retrieval

Input: a query image i, dense text set:$\{t_n^d\}_{n=1}^k$, sparse text set:$\{t_n^p\}_{n=1}^k$
Parameter: δ, φ, μ
Output: Selected set A

```
Calculate normalized ranking dense scores D and sparse scores S.
Select top-k sparse vectors from S.
Perform K-means clustering on D and save the clusters offline.
Initialize empty set A.
for each cluster C containing top-δk sparse vectors do
    Calculate priority score W(T_i) for cluster C.
    if W(T_i) > φ then
        Add cluster C to set A.
    end if
end for
Prune set A: let |A| ≤ μk.
return set A
```

Estimate the mean and variance of these scores assuming they follow a normal distribution:

$$\mu_k = \frac{1}{m}\sum_{i=1}^{m} S(I_i, T_{i,j(k)}) \tag{16}$$

$$\sigma_k = \sqrt{\frac{\sum_{i=1}^{m}\left(S(I_i, T_{i,j(k)}) - \mu_k\right)^2}{m}} \tag{17}$$

$$\lim_{m\to\infty} \Pr\left(\frac{S_{(k)} - \mu_k}{\sigma_k} \le Z_\epsilon\right) = \epsilon. \tag{18}$$

Thus, given ϵ, find Z_ϵ by standard statistical methods and compute an unbiased estimate of ϕ_k: $\phi_k = \mu_k + Z_\epsilon\sigma_k$. Therefore, for each text cluster, we calculate its lower bound φ when it contains at least one of the top θk texts

$$W(T_i) \ge \frac{\phi_{\theta k}}{\log(\theta k + 1)} = \varphi. \tag{19}$$

Then, we construct the set A_c

$$A_c = \{T_c | (T_c \cap \mathrm{TOPS}(\delta k)) \cup W(T_c) \ge \varphi\}. \tag{20}$$

Finally, we further refine the set A_c by pruning, ensuring that $|A_c| \le \mu k$, thereby further limiting the size of the set and saving subsequent computation space. Compared to directly using the current δk sparse results to guide dense retrieval, our approach can recognize results with more diverse structures. Therefore, we only need to calculate the similarity between the corresponding few dense vectors in set A_c and the sparse vectors, and perform linear interpolation to obtain the final retrieval score for image I_i. We estimate the threshold value φ for priority scoring by calculating the minimum score that approximately includes the top θk sparse results within a cluster. This process leverages the asymptotic normal distribution properties of order statistics. For instance, in the image-to-text retrieval task, since retrieval is performed on the entire text dataset, this asymptotic property can be applied to the Kth retrieval result for each image query.

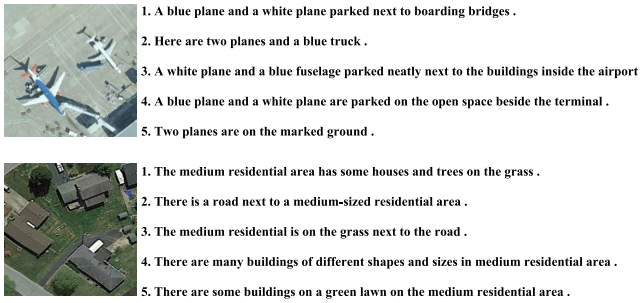


Fig. 3. Two examples from the CMRSITR dataset, where each RS image is associated with five captions.

IV. Experiments and Discussion

A. Dataset and Evaluation Metrics

1) Datasets:

a) Small-scale datasets: To evaluate the effectiveness of our SGPD framework, we conducted extensive experiments on several small-scale RS image–text datasets: RSICD [7], RSITMD [8], and UCM [20]. As shown in Fig. 3, each sample in the above dataset contains an RS image and five corresponding captions. RSICD is currently the largest and most widely used RSITR dataset, containing 10 921 RS images, each with a fixed size of 224 × 224 pixels, and five descriptive sentences associated with each image. RSITMD is a condensed version of RSICD, comprising RS images sourced from both RSICD and Google Earth. It contains a total of 4743 images and 23 715 descriptive sentences. UCM, derived from the UC Merced land use dataset, includes 2100 images, with each image paired with five manually annotated descriptive texts.

b) Large-scale datasets: To evaluate the effectiveness of our method on large-scale datasets, we integrated the test sets from RSICD, RSITMD, and NWPU, along with 1 million image–text pairs randomly selected from RS5M [54]. For convenience, we refer to the combined dataset of RSICD, RSITMD, and NWPU as SET3, and the extended dataset, which includes the additional 1 million image–text pairs from RS5M, as large-scale SET3. In the text-to-image retrieval task, the model uses 23 475 captions from SET3 as queries and retrieves results from 4695 images in SET3 and 1 million images from RS5M. Similarly, in the image-to-text retrieval task, the model uses 4695 images from SET3 as queries and retrieves results from 23 475 captions in SET3 and 1 million captions from RS5M.

2) Evaluation Metrics: To evaluate the effectiveness of our SGPD method, we selected two widely used evaluation metrics: R@k (k = 1, 5, 10) and mR. R@k represents the percentage of correctly matched pairs among the top K retrieval results, while mR is the mean of R@k values across all retrievals.

B. Implementation Details

Our model uses pretrained weights from BLIP, which were trained on the MSCOCO [55], and finetunes them on the SET3 dataset for 10 epochs across three RTX 3090 GPUs to generate precomputed dense vectors for RS images and captions. Next, we train the model to convert the dense vectors into sparse vectors for 150 epochs. The temperature parameter ($\mathcal{T}$) is set to 0.001, and the sparse regularization weight (η) is set to e^{-5}.

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To validate the effectiveness of our sparsification strategy, we first evaluate the performance using only the sparse vectors on the small-scale datasets.

C. Comparison With State-of-the-Art Methods

In our experimental study on small-scale datasets, we conducted a comprehensive comparative analysis. Since there is currently no sparse retrieval paradigm specifically designed for RSITR tasks, we selected representative traditional methods (VSE++ [56], SCAN [57], AMFMN [8], HVSA [58], IEFT [59]), as well as VLP-based cross-modal retrieval algorithms (PIR [60], PE-RSITR [10], MIIA [9], RemoteCLIP [11], GLISA [61], IERR [62], and GeoRSCLIP [54]). It is important to note that all of the aforementioned methods are based on the dense retrieval paradigm.

1) *VSE++ [56]:* VSE++ introduces contrastive loss and an improved triplet loss, enhancing the matching accuracy between images and texts through hard negative mining.
2) *SCAN [57]:* SCAN introduces a stacked cross-attention model to achieve effective image–text matching, by mining the intrinsic relationships between specific regions in the image and the corresponding words in the text.
3) *AMFMN [8]:* MFMN introduces adaptive multiscale feature inputs and dynamically filters redundant features, using a multiscale visual self-attention module to extract salient features from RS images.
4) *HVSA [58]:* HVSA introduces a hypersphere-based visual-semantic alignment network, which treats sample pairs of varying difficulty differently. It distinguishes samples based on their complexity, leading to improved embedding representations.
5) *IEFT [59]:* IEFT introduces a simplified feature embedding module to map images and texts into distinct visual and textual feature spaces. It also incorporates an information interaction enhancement module, which simultaneously models the intrinsic relationships between RS images and texts.
6) *PIR [60]:* PIR introduces a paradigm that utilizes prior knowledge to guide the adaptive learning of visual and textual representations. It filters key features through the establishment of a belief matrix, aiming to address the semantic noise issue in visual-language understanding tasks within the RS domain.
7) *PE-RSITR [10]:* PE-RSITR utilizes the CLIP pretrained model and introduces a PETL approach to enhance performance. It leverages a multimodal RS adapter and a hybrid multimodal contrastive learning objective to transfer visual-language knowledge from the natural domain to the RS domain.
8) *MIIA [9]:* MIIA introduces a masked interaction inference mechanism based on a pretrained VL model. It generates two dynamically updated negative sample queues, allowing the model to learn more visual-textual features.
9) *RemoteCLIP [11]:* The first VL foundation model designed for RS, which learns visually rich semantic features and aligns them with textual embeddings. It enables a variety of downstream tasks, including zero-shot image classification, image–text retrieval, and few-shot classification.
10) *GLISA [61]:* GLISA presents a transformer-based CMRSITR framework, enhancing retrieval performance with global–local information soft alignment (GLISA). The framework includes a global image extraction module and an adaptive local information extraction (ALIE) module, while the local information soft alignment (LISA) module alleviates semantic ambiguity for improved alignment and retrieval.
11) *IERR [62]:* IERR proposes a RSITR method based on implicit–explicit relationship reasoning (IERR). By integrating masked image modeling (MIM) and masked language modeling (MLM), the method achieves symmetric mask reasoning consistency alignment, addressing the visual bias caused by perceptual variations and the geographic semantic confusion due to spatial proximity in RS images.
12) *GeoRSCLIP [54]:* Introduces RS5M, the largest image–text paired dataset to date, consisting of 5 million RS images and their corresponding English captions. CLIP is fine-tuned on this dataset to perform RSITR tasks.

Extensive comparative experiments on small-scale datasets such as RSICD, RSITMD, and UCM (see Tables I–III) lead to the conclusion that VLP methods generally outperform traditional cross-modal retrieval methods like VSE++ and HVSA. Notably, the proposed SGPD framework demonstrates state-of-the-art performance across all benchmarks. On the RSICD and RSITMD datasets, SGPD (BLIP) significantly outperforms existing methods. For instance, on RSICD, particularly in the image-to-text task, SGPD (BLIP) achieves R@1 (21.86%) and R@5 (43.82%), surpassing the top two methods, GeoRSCLIP and RemoteCLIP, by 4.73% and 2.1%, respectively. Additionally, on the RSITMD dataset, SGPD (BLIP) achieves a mean retrieval rate (mR) of 53.78%, outperforming GeoRSCLIP (51.81%) and RemoteCLIP (49.38%) by 1.97% and 4.4%, respectively.

The advantages of VLP methods are further validated by zero-shot ALBEF and zero-shot BLIP. In the text-to-image retrieval task on RSICD, zero-shot BLIP achieves an R@1 of 10.67%, outperforming the traditional method IEFT (8.38%) by 27%. On the UCM dataset, SGPD sets new benchmarks in both image-to-text (R@1 = 25.71%) and text-to-image (R@10 = 93.52%) tasks, with the highest mean retrieval rate (mR) of 58.09%. Compared to the second-best method, PE-RSITR, SGPD improves performance by 4.38% in image-to-text and 1.73% in text-to-image tasks.

Moreover, the multimodal integration strategy within the SGPD framework, incorporating ALBEF and BLIP variants, provides flexibility for various application scenarios and demonstrates strong cross-modal alignment capabilities across multiple datasets. The extensive experimental results not only validate the effectiveness of SGPD on small-scale datasets but also lay a solid foundation for efficient retrieval on large-scale datasets in the future.

D. Ablation Study

To validate the effectiveness of our method on small-scale datasets, we conducted detailed ablation experiments. First,

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TABLE I
PERFORMANCE COMPARISONS WITH STATE-OF-THE-ART METHODS ON RSICD DATASET

Method	Training Dataset	Image Backbone	Text Backbone	Params	Image to Text			Text to Image			mR
					R@1	R@5	R@10	R@1	R@5	R@10	
VSE++ [56]	RSICD	ResNet-18	GRU	15.78M	3.38	9.51	17.46	2.82	11.32	18.10	10.43
SCAN [57]	RSICD	Faster R-CNN	BiGRU	13.68M	5.85	12.89	19.84	3.71	16.40	26.73	14.23
AMFMN [8]	RSICD	Faster R-CNN	BiGRU	35.94M	5.39	15.08	23.40	4.90	18.28	31.44	16.42
HVSA [58]	RSICD	Faster R-CNN	BiGRU	—	7.47	20.62	32.11	5.51	21.13	34.13	16.43
IEFT [59]	RSICD	ViT	BERT	—	8.78	28.47	43.88	8.38	28.17	44.16	26.97
PIR [60]	RSICD	Swin-T+ResNet-50	BERT	—	9.88	27.26	39.16	6.97	24.56	38.92	24.46
PE-RSITR [10]	RSICD	ViT-B/32	Transformer	0.16M	14.13	31.51	44.78	11.63	33.92	50.73	31.12
MIIA [9]	RET-3	ViT-B/16	Transformer	—	11.05	30.17	49.32	13.21	34.74	45.60	30.68
RemoteCLIP [11]	RemoteCLIP Datasets	ResNet-50	Transformer	102M	13.36	32.94	44.83	10.76	32.83	48.75	30.58
RemoteCLIP [11]	RemoteCLIP Datasets	ViT-B/32	Transformer	151M	17.02	37.97	51.51	13.71	37.11	54.25	35.26
GLISA [61]	RSICD	ViT	Transformer	—	19.52	40.44	52.28	14.75	39.50	55.46	36.99
IERR [62]	RSICD	ViT	Transformer	—	17.29	35.41	48.58	13.07	35.32	51.31	33.50
GeoRSCLIP [54]	RS5M	ViT-B/32	Transformer	—	21.13	41.72	55.63	15.59	41.19	57.99	38.87
Zero-Shot ALBEF [51]	—	ResNet-50	BERT	0M	7.82	19.34	30.41	6.18	20.35	33.45	19.59
Zero-Shot BLIP [52]	—	ViT-B/16	Transformer	0M	9.15	22.96	32.02	10.67	27.75	40.00	23.76
SGPD(ALBEF)	SET3	ResNet-50	BERT	7.88M	16.74	34.15	47.12	11.93	34.61	49.61	32.36
SGPD(BLIP)	SET3	ViT-B/16	Transformer	7.88M	**21.86**	**43.82**	**57.09**	**18.20**	**43.91**	**60.14**	**40.83**

TABLE II
PERFORMANCE COMPARISONS WITH STATE-OF-THE-ART METHODS ON RSITMD DATASET

Method	Training Dataset	Image Backbone	Text Backbone	Params	Image to Text			Text to Image			mR
					R@1	R@5	R@10	R@1	R@5	R@10	
VSE++ [56]	RSITMD	ResNet-18	GRU	15.78M	10.38	27.65	39.60	7.79	24.87	38.67	24.83
SCAN [57]	RSIMD	Faster R-CNN	BiGRU	13.68M	11.06	25.88	39.38	9.82	29.38	42.12	26.28
AMFMN [8]	RSITMD	Faster R-CNN	BiGRU	35.94M	10.63	24.78	41.81	11.51	34.69	54.87	29.72
HVSA [58]	RSITMD	Faster R-CNN	BiGRU	—	13.20	32.08	45.58	11.43	39.20	57.45	33.16
IEFT [59]	RSITMD	ViT	BERT	—	15.49	37.61	51.40	11.19	38.09	58.84	35.44
PIR [60]	RSITMD	Swin-T+ResNet-50	BERT	—	18.14	41.15	52.88	12.17	41.68	63.41	38.24
PE-RSITR [10]	RSITMD	ViT-B/32	Transformer	0.16M	23.67	44.07	60.36	20.10	50.63	67.97	44.47
MIIA [9]	RET-3	ViT-B/16	Transformer	—	19.82	52.10	68.11	23.41	45.21	63.29	45.32
RemoteCLIP [11]	RemoteCLIP Datasets	ResNet-50	Transformer	102M	23.67	47.57	64.60	19.29	51.55	70.58	46.21
RemoteCLIP [11]	RemoteCLIP Datasets	ViT-B/32	Transformer	151M	27.88	50.66	65.71	22.17	56.46	73.41	49.38
GLISA [61]	RSITMD	ViT	Transformer	—	28.21	50.94	62.50	20.50	54.80	72.19	48.41
IERR [62]	RSITMD	ViT	Transformer	—	27.66	48.89	61.06	24.43	56.59	72.66	48.55
GeoRSCLIP [54]	RS5M	ViT-B/32	Transformer	—	32.30	53.32	67.92	25.04	58.88	74.38	51.81
Zero-Shot ALBEF [51]	—	ResNet-50	BERT	0M	12.45	23.57	32.45	13.41	33.92	47.71	27.25
Zero-Shot BLIP [52]	—	ViT-B/16	Transformer	0M	14.38	26.99	37.61	16.50	38.58	50.61	30.78
SGPD(ALBEF)	SET3	ResNet-50	BERT	7.88M	25.14	48.04	63.61	20.33	53.93	72.56	47.26
SGPD(BLIP)	SET3	ViT-B/16	Transformer	7.88M	**33.21**	**53.41**	**70.21**	**27.03**	**59.41**	**79.39**	**53.78**

TABLE III
PERFORMANCE COMPARISONS WITH STATE-OF-THE-ART METHODS ON UCM DATASET

Method	Training Dataset	Image Backbone	Text Backbone	Params	Image to Text			Text to Image			mR
					R@1	R@5	R@10	R@1	R@5	R@10	
VSE++ [56]	UCM	ResNet-18	GRU	15.78M	12.38	44.76	65.71	10.10	31.80	56.85	36.93
SCAN	UCM	Faster R-CNN	BiGRU	13.68M	12.85	47.14	69.52	12.48	46.86	71.71	43.43
AMFMN [8]	UCM	Faster R-CNN	BiGRU	35.94M	16.67	45.71	68.57	12.86	53.24	79.43	46.08
PE-RSITR [10]	UCM	ViT-B/32	Transformer	0.16M	22.71	55.81	80.33	**18.82**	**62.84**	93.72	55.71
MIIA [9]	RET-3	ViT-B/16	Transformer	—	15.69	51.07	75.21	15.13	60.21	80.97	49.71
RemoteCLIP [11]	RemoteCLIP Datasets	ResNet-50	Transformer	102M	13.33	50.48	74.76	15.24	57.14	84.57	49.25
RemoteCLIP [11]	RemoteCLIP Datasets	ViT-B/32	Transformer	151M	20.48	59.85	83.33	18.67	61.52	**94.29**	56.36
Zero-Shot ALBEF [51]	—	ResNet-50	BERT	0M	10.31	37.12	58.31	9.56	32.68	55.94	33.98
Zero-Shot BLIP [52]	—	ViT-B/16	Transformer	0M	13.33	40.47	60.47	13.90	50.09	75.71	42.23
SGPD(ALBEF)	SET3	ResNet-50	BERT	7.88M	15.73	52.46	73.45	14.78	59.13	79.18	49.12
SGPD	SET3	ViT-B/16	Transformer	7.88M	**25.71**	**62.85**	**88.57**	15.61	62.28	93.52	**58.09**

we explored the impact of different sparse regularization weights on the model's performance to understand how varying the regularization parameter affects the balance between sparsity and retrieval accuracy. Next, we investigated the effectiveness of each component within the SGPD framework. By systematically removing or modifying specific components, we gained valuable insights into how each part contributes to the overall model's performance. These experiments helped identify the most influential elements of our model, providing valuable guidance for further optimization and refinement.

1) Sparse Regularization Weights: To evaluate the effect of sparse regularization weights η, we conducted retrieval experiments using only sparse vectors on small-scale datasets. We tested a range of sparse regularization weights, varying from $1 \times e^{-5}$ to $2 \times e^{-2}$. Alongside retrieval performance, we introduced FLOPs as a metric to assess the model's computational efficiency

$$\text{FLOPs} = \frac{1}{|T||I|} \sum_{X_T \in T} \sum_{X_I \in I} S_T^0 S_I^0 \tag{21}$$

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TABLE IV
IMPACT OF DIFFERENT SPARSE REGULARIZATION WEIGHTS ON DIFFERENT DATASETS WITH ALBEF

Method	Image to Text			Text to Image			
	R@1	R@5	R@10	R@1	R@5	R@10	FLOPs
RSICD Dataset							
Fine-tuning ALBEF	**18.14**	**36.13**	**50.31**	**14.18**	**35.13**	**53.51**	256.00
SGPD($\eta = 2e-2$)	12.98	28.41	39.13	7.38	25.12	35.24	**3.51**
SGPD($\eta = 1e-3$)	14.01	30.74	38.51	8.33	28.45	38.67	14.13
SGPD($\eta = 1e-4$)	14.65	31.73	43.65	8.95	30.78	40.89	19.56
SGPD($\eta = 1e-5$)	16.74	34.15	47.12	11.93	34.61	49.61	20.14
UCM Dataset							
Fine-tuning ALBEF	**17.31**	**55.72**	**76.74**	**18.43**	**62.34**	**83.96**	256.00
SGPD($\eta = 2e-2$)	10.03	40.12	55.23	10.05	45.34	60.45	**2.84**
SGPD($\eta = 1e-3$)	11.51	45.23	60.34	12.07	50.56	65.67	16.31
SGPD($\eta = 1e-4$)	13.15	50.34	65.45	13.09	55.78	70.89	23.45
SGPD($\eta = 1e-5$)	15.73	52.46	73.45	14.78	59.13	79.18	24.25

TABLE V
IMPACT OF DIFFERENT SPARSE REGULARIZATION WEIGHTS ON DIFFERENT DATASETS WITH BLIP

Method	Image to Text			Text to Image			
	R@1	R@5	R@10	R@1	R@5	R@10	FLOPs
RSICD Dataset							
Fine-tuning BLIP	**25.25**	**44.92**	**59.28**	**20.14**	**47.48**	**63.31**	256.00
SGPD($\eta = 2e-2$)	17.65	36.96	50.77	13.52	36.65	52.75	**2.76**
SGPD($\eta = 1e-3$)	20.58	43.09	56.17	16.87	44.20	60.25	15.01
SGPD($\eta = 1e-4$)	21.68	43.45	58.73	18.02	44.51	61.00	29.90
SGPD($\eta = 1e-5$)	21.86	43.82	57.09	18.20	43.91	60.14	25.47
UCM Dataset							
Fine-tuning BLIP	**25.71**	**62.85**	**88.57**	**15.61**	**62.28**	**93.52**	256.00
SGPD($\eta = 2e-2$)	18.09	53.80	83.80	14.85	55.80	85.71	**2.02**
SGPD($\eta = 1e-3$)	18.57	57.21	86.66	14.38	58.19	89.24	17.91
SGPD($\eta = 1e-4$)	18.09	57.14	86.66	16.47	58.66	88.28	37.21
SGPD($\eta = 1e-5$)	20.00	58.57	87.61	15.61	61.04	90.90	34.73

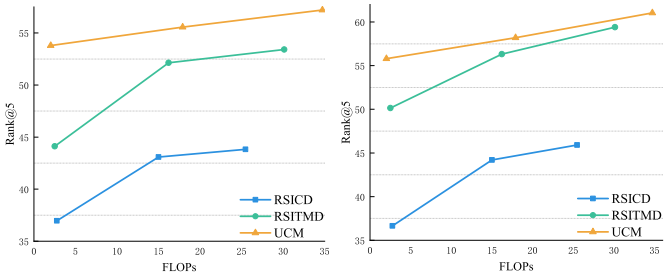


Fig. 4. Performance comparison of different sparse regularization weights on different datasets. (a) Image to text R@5. (b) Text to image R@5.

where $|T|$ and $|I|$ are text and image collections, and S_T and S_I are sparse embeddings of a text X_T and an image X_I.

Fig. 4 illustrates the trade-off between effectiveness and efficiency as sparse regularization weights vary. Experiments conducted on three small-scale datasets show that increasing the sparse regularization weight reduces FLOPs, thereby improving the model's computational efficiency. However, this improvement comes at the cost of reduced recall and retrieval effectiveness.

As shown in Tables IV and V, we experimentally analyze the impact of sparse regularization weights on the performance of the BLIP and ALBEF models, and investigate the sensitivity of regularization parameters across different datasets (RSICD and UCM). Various sparse regularization weights (η ranging from $1 \times e^{-5}$ to $2 \times e^{-2}$) are tested, with performance evaluated using image-to-text and text-to-image recall rates (R@1, R@5, R@10) and computational efficiency (FLOPs).

The results show that the choice of regularization parameter plays a crucial role in balancing retrieval accuracy and computational efficiency. For both the BLIP and ALBEF models, the optimal balance is achieved when the regularization weight is set to $\eta = 1 \times e^{-5}$. For example, on the RSICD dataset, the BLIP model achieves an image-to-text R@1 of 21.86%, a text-to-image R@1 of 14.82%, and a FLOPs of 15.01, significantly reducing computational cost compared to the dense model (FLOPs = 256). Similarly, with ALBEF at $\eta = 1 \times e^{-5}$, the image-to-text R@1 is 16.74%, the text-to-image R@1 is 11.93%, and FLOPs is 19.56, demonstrating that a smaller regularization weight not only improves retrieval performance but also maintains high computational efficiency.

As the regularization weight increases (e.g., $\eta = 2 \times e^{-2}$), we observe improved computational efficiency, with FLOPs decreasing to 2.76 (BLIP) and 3.51 (ALBEF), but retrieval accuracy significantly decreases. For instance, with $\eta = 2 \times e^{-2}$, the BLIP model on RSICD shows an image-to-text R@1 of 17.65% and a text-to-image R@1 of 13.52%, indicating that excessively large regularization weights improve efficiency at the cost of retrieval accuracy.

2) Learning Sparse Representations: In Table VI, we present a comprehensive ablation study to analyze the impact of different training objectives and stages on model performance using sparse vectors on small-scale datasets. The components analyzed include SP, ℓ_1 regularization (LR), and FR.

The second row of Table VI shows the results after adding SP to the baseline. Following the introduction of SP, the model's performance on the RSICD dataset experiences a slight drop: the text-to-image R@1 decreases to 13.52%, and the image-to-text R@1 decreases to 17.42%. However, the computational cost significantly reduces, with FLOPs dropping to 84.23. This suggests that SP simplifies the computation but may somewhat limit the model's expressiveness. After adding LR, the model's sparsity is further enhanced, leading to more weights tending toward zero, which effectively simplifies the parameter space. On the RSICD dataset, text-to-image R@1 improves slightly to 16.11%, and image-to-text R@1 increases to 18.35%, with FLOPs further decreasing to 43.31. A similar trend is observed on the UCM dataset, where both performance and computational efficiency improve. When all three components are combined, as shown in the fifth row, the model achieves the best balance between performance and efficiency. On the RSICD dataset, text-to-image R@1 reaches 18.20%, image-to-text R@1 reaches 21.86%, while maintaining the lowest computational cost at 25.47 FLOPs. On the UCM dataset, text-to-image R@1 reaches 15.61%, image-to-text R@1 reaches 18.57%, and FLOPs decrease to 34.73. This demonstrates that the combination of SP, LR, and FR has a synergistic effect, enabling the generation of both efficient and effective feature representation models.

E. Retrieval Performance on Large-Scale Datasets

1) Implementation Details: We use the SET3 training dataset and exclude the large-scale Set3 dataset from

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TABLE VI
ABLATION STUDY ON EACH COMPONENT OF SGPD ON RSICD AND UCM DATASETS

No.	Components			RSICD							UCM						
				Text to Image			Image to Text			FLOPs	Text to Image			Image to Text			FLOPs
	SP	LR	FR	R@1	R@5	R@10	R@1	R@5	R@10		R@1	R@5	R@10	R@1	R@5	R@10	
1	✓			13.52	33.41	53.31	17.42	34.61	52.51	84.23	9.16	52.83	82.15	13.47	51.33	80.31	68.34
2	✓	✓		16.11	39.51	56.33	17.71	38.31	55.12	43.31	13.71	59.31	87.31	14.84	55.12	85.12	52.51
3	✓		✓	15.27	40.31	58.29	18.41	41.23	56.31	**25.33**	14.32	58.71	88.12	16.39	56.55	84.23	39.13
4	✓	✓	✓	**18.20**	**43.91**	**60.14**	**21.86**	**43.82**	**57.09**	25.47	**15.61**	**61.04**	**89.90**	**18.57**	**57.21**	**87.61**	**34.73**

TABLE VII
RETRIEVAL PERFORMANCE OF OUR SGPD AND BASELINES IN LARGE-SCALE SET3 TEST

Method	Cand.	QPS	Time	Space	T2I	I2T
ALBEF	1M	2.28	213.77ms	26,122M	28.33	31.07
BLIP	1M	2.31	228.30ms	26,122M	30.10	32.53
BM25	1M	513.37	3.68ms	**268M**	10.43	13.14
SPLADE	1M	**556.66**	**2.95ms**	332M	21.46	19.33
IVF	**0.3K**	371.36	9.12ms	230M	20.93	22.15
SGPD(ALBEF)	**0.3K**	524.15	5.83ms	452M	23.44	28.31
SGPD(BLIP)	**0.3K**	503.47	6.17ms	473M	**29.37**	**30.62**

the RS5M for pretraining. For single-modal text retrieval, we employ BM25 and SPLADE, where text captions substitute for images in the image–text retrieval task. Sparse retrieval is performed using the Anserini library [63], which efficiently conducts similarity searches with sparse vectors. However, during the inference phase of large-scale retrieval, a notable difference arises between dense and sparse retrieval methods. In our neural network, we use the dot product of sparse dictionary-weighted representations to measure similarity, which is essential for gradient backpropagation. However, this approach is not feasible for term-based sparse retrieval systems, such as Anserini. To overcome this limitation, we adopt a quantization method that transforms high-dimensional sparse vectors into corresponding dictionaries and their virtual weights. These dictionaries are derived from the nonzero elements of the high-dimensional sparse vectors, while their weights are determined through a straightforward quantization procedure. Given a query and a candidate sample, the dictionary matching score can be defined as follows:

$$\text{Score} = \sum_{l \in L^q \cap L^c} W_q(h) \times W_c(h) \tag{22}$$

where L^q and L^c represent the dictionary lists for the query and candidate sample, respectively, and $W_{(\cdot)}(l)$ denotes the weight of the dictionary.

To interpolate between sparse and dense models, we apply min–max normalization to rescale the top results for each query. All models are initially embedded into index files, with retrieval conducted on a server configured with an Intel i7-1260P CPU, 64 GB of memory, and a PCIe SSD. Dense vectors are clustered using the K-means algorithm from the Faiss library [64], with a fixed cluster size of 126, and the number of clusters is adjusted as necessary.

2) Main Results: As shown in Table VII, SGPD (BLIP) achieves an R@10 score of 29.37% for text-to-image retrieval and 30.62% for image-to-text retrieval. Compared to the original BLIP baseline (T2I: 30.10%, I2T: 32.53%), the accuracy drop is relatively small, with T2I decreasing by 0.73 percentage points and I2T decreasing by 1.91 percentage points. Additionally, SGPD (ALBEF) achieves T2I and I2T scores of 23.44% and 28.31%, respectively, which are slightly lower than the ALBEF baseline (T2I: 28.33%, I2T: 31.07%). However, SGPD (ALBEF) significantly improves retrieval efficiency by compressing the candidate set size (from 1 M to 0.3 k). In large-scale retrieval scenarios, SGPD methods—especially SGPD (BLIP)—outperform traditional sparse models (BM25, SPLADE) and dense models (IVF), demonstrating their strong cross-modal representation capabilities. For example, compared to IVF, SGPD (BLIP) improves T2I and I2T accuracy by 8.44% and 8.47%, respectively, highlighting its advantages in dense feature clustering and cross-modal mapping.

In large-scale retrieval tasks, query-per-second (QPS) and storage overhead are critical metrics for practical deployment. As shown in Table VII, SGPD (BLIP) achieves a retrieval time of 6.17 ms (QPS: 503.47), significantly outperforming the original BLIP (228.30 ms, QPS: 2.31) and ALBEF (213.77 ms, QPS: 2.28). SGPD (ALBEF) achieves even higher retrieval efficiency (5.83 ms, QPS: 524.15), surpassing IVF (9.12 ms, QPS: 371.36). This improvement is attributed to SGPD's optimized dense vector indexing and cross-modal feature fusion mechanism.

Although SGPD (BLIP) requires more storage (473 MB) than sparse methods like BM25 (268 MB) and dense models like IVF (230 MB), its exceptional retrieval accuracy (T2I: 29.37%) and efficiency (QPS: 500+) in large-scale tasks justify the overhead. SGPD (ALBEF) has slightly lower storage requirements (452 MB), further balancing performance and resource consumption. Notably, SGPD methods reduce the candidate set size from 1 M in the baseline models to 0.3 k, significantly lowering computational complexity. For example, while BLIP and ALBEF use 1 M candidate sets, SGPD variants only require 0.3 k. This compression, achieved through the proposed clustering algorithm and multimodal feature fusion, not only accelerates retrieval speed (QPS increased to 500+) but also reduces resource usage, making SGPD more suitable for practical deployment scenarios.

3) Ablation Study: Table VIII presents the results of the ablation study on the large-scale SET3 dataset, where "Clu" refers to the number of selected embedding clusters, with the retrieval depth fixed at 1000 ($k = 1000$). As the values of δ and θ increase, the model's retrieval performance gradually improves. This suggests that in the SGPD model, larger probability control parameters help capture richer semantic

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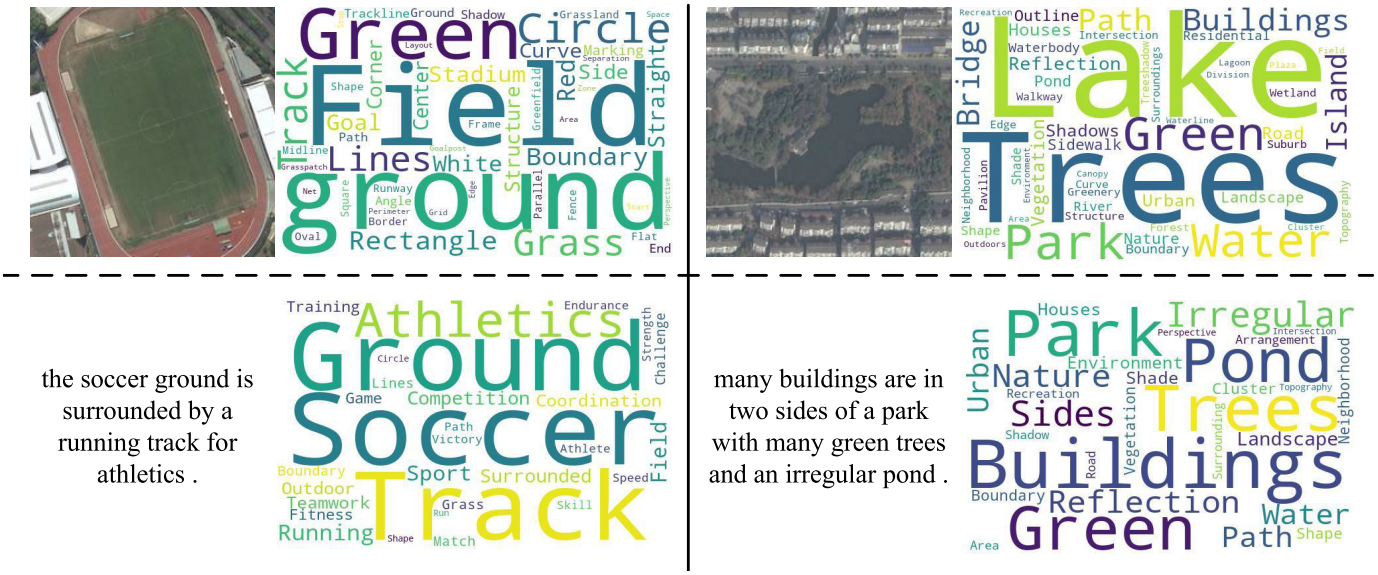


Fig. 5. Visualization of the word clouds for images in the SET3 test set and their corresponding captions. The top 50 predicted tokens, extracted from either the image or the caption, are shown, with font size indicating the predicted weight of each token.

TABLE VIII
PERFORMANCE WITH DIFFERENT VARIABLE ON LARGE-SCALE DATASET

Para	I2t Retrieval R@5	I2t Retrieval R@10	T2I Retrieval R@5	T2I Retrieval R@10	Clu	Time/ms
(δ,θ), μ =0.02						
(0.01,0)	8.13	17.32	7.11	17.26	**4.20**	**3.24**
(0.02,0)	9.31	20.14	8.23	23.51	7.10	4.33
(0,0.02)	11.04	25.12	10.31	26.12	10.80	5.78
(0.02,0.02)	**11.35**	**27.31**	**12.01**	**28.41**	12.50	5.46
(δ,θ), μ =0.03						
(0.01,0)	10.32	26.13	9.25	25.12	**6.10**	**3.79**
(0.02,0)	11.44	28.23	12.44	28.41	8.50	5.03
(0,0.02)	12.03	28.16	11.78	30.13	13.60	6.03
(0.02,0.02)	**12.51**	**29.30**	**13.15**	**30.62**	14.30	6.17

information, thereby improving matching accuracy. Additionally, the number of selected embedding clusters increases as δ and θ grow, indicating that the model considers a broader range of potential sparse features, which enhances its generalization ability. However, the rapid growth of "Clu" may lead to increased computational overhead, requiring a trade-off.

As the values of δ and θ increase, the model can handle more sparse features, leading to more precise matching in both I2T and T2I retrieval tasks. Specifically, when $\delta = \theta = 0.02$, the R@5 and R@10 metrics reach their optimal performance, indicating that these hyperparameters help to better exploit sparse features. However, excessively large values of δ and θ may significantly increase inference time, necessitating careful control of their growth to prevent excessive computational costs.

On the other hand, the hyperparameter μ primarily influences the number of clusters. Increasing μ improves the model's matching accuracy, particularly when $\mu = 0.03$, where both R@5 and R@10 metrics show substantial improvements. A larger number of clusters allows the model to cover more candidate samples, thereby enhancing retrieval comprehensiveness and accuracy. However, the increased number of clusters also introduces additional computational overhead, so when optimizing hyperparameters, it is essential to strike a balance between performance and computational efficiency.

F. Qualitative Results

Fig. 5 presents the word cloud visualizations for two RS images and their corresponding captions. The size of the words in the word cloud is directly proportional to their weight, reflecting the model's emphasis on key features in the RSITR task. Words with higher weights appear in larger fonts, highlighting the visual and semantic features that the model deems most important during the image–text matching process.

From the word cloud of the first image, it is evident that the model successfully captures the main features associated with both the image and its textual caption. The word "ground" is notably enlarged, aligning with the visual features of the sports field in the image and the text caption, "the soccer ground is surrounded by a running track for athletics." Other words, such as "track" and "field," further suggest that the model effectively identifies details related to the field and track, retrieving the corresponding text captions. In the second image and its word cloud, a similar pattern emerges. Keywords like "lake," "trees," and "buildings" occupy prominent positions in the word cloud, accurately reflecting the semantic features of the main scene elements in the image—such as the lake, greenery, and buildings. This prominence aligns with the description in the caption, "many buildings are on two sides of a park with many green trees and an irregular pond," further validating the model's ability to extract semantic features from the image and retrieve the relevant captions from the text.

Fig. 6 presents the qualitative results of our method in the RSITR task. Given a query, the top five retrieval results are arranged from left to right. Correct results are highlighted in

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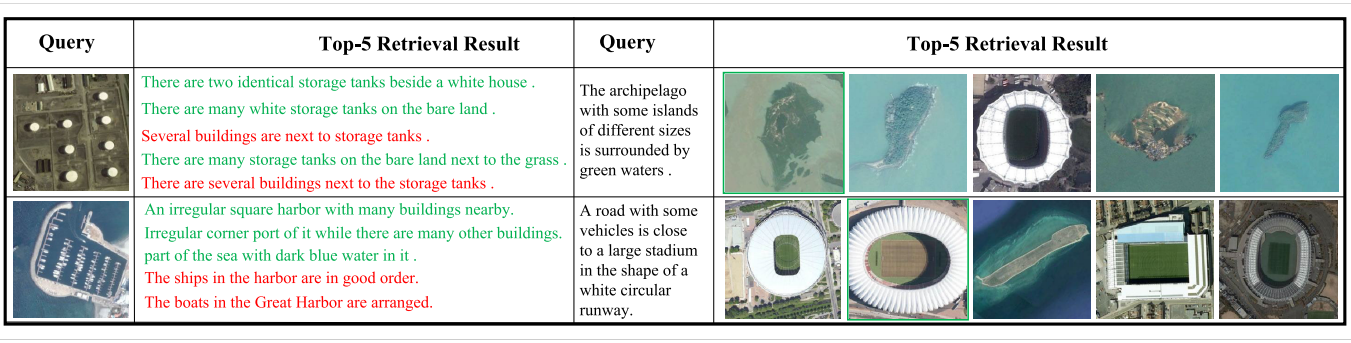


Fig. 6. Given a query image or text from the SET3 test set, the top five retrieval results are listed. Correct retrieval results are marked in green, while incorrect results are marked in red.

green, while incorrect results are marked in red. It can be observed that our method retrieves accurate matching results in most cases, positioning the correct results at the top even in complex scenes, such as those involving "white storage tanks" and "harbor" in the image. This demonstrates the model's robustness in capturing the core semantic matching between visual features and text captions.

However, in some cases, results marked as "incorrect" could still represent reasonable captions. For example, in the retrieval results for the first image, the caption "Several buildings are next to storage tanks" is marked as incorrect but aligns well with the image content. This discrepancy may stem from limitations in semantic annotation within the dataset, where some captions are exclusively labeled as titles for other images, leading to a reasonable retrieval being mistakenly classified as incorrect. In multimodal retrieval of RS images, the limited annotation resources and diverse scene captions imply that a single title may not encompass all possible semantic interpretations of an image, particularly in cases where semantic features overlap.

V. Conclusion and Future Work

In this study, we propose a novel RSITR method, SGPD Retrieval. This approach effectively combines the strengths of sparse and dense retrieval by leveraging initial scores from sparse retrieval to perform more precise searches within a smaller set of dense vectors. SPGD is designed for seamless integration into existing image–text retrieval models without requiring modifications to their original architecture. Experimental results demonstrate that SPGD achieves both efficient and accurate retrieval performance across small-scale and large-scale datasets. Furthermore, the method enhances the performance of image–text retrieval models in cross-modal tasks, offering more generalized retrieval capabilities.

Since SGPD depends on existing pretrained VL models, its retrieval accuracy is inherently constrained by the performance of the underlying models. Additionally, current RS image–text datasets often contain overly simplistic and singular text captions, which further limit the model's training and inference capabilities. Future work will focus on developing large-scale RS image–text datasets with more accurate and detailed semantic annotations, aiming to significantly advance the progress of RSITR technology.

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